

GMaster: a difference-form Wigner- d march for GPU MASTER estimation

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ABSTRACT

Estimating binned angular power spectra from masked high-resolution full-sky maps—through spherical harmonic transforms and pseudo- C_ℓ (MASTER) estimators that propagate survey windows into mode coupling—remains a practical bottleneck in CMB and large-scale structure pipelines. On GPUs the latitudinal factor of a HEALPix transform is often evaluated as a serial Jacobi recurrence in degree; in float32 that recurrence is unstable near the poles and at turning points, where a rounding of size $u|v|$ is amplified by $1/\sin\phi$. We rewrite the march so that the carried pair is (v_n, D_n) with $D_n = v_n - v_{n-1}$ rather than (v_n, v_{n-1}) . The difference coefficient vanishes as ϕ^2 exactly where the plain form is ill-conditioned, error is injected at the scale of $D \sim \phi v$, and the float32 floor becomes the random walk $\sqrt{n}u$. Against the float64 recurrence at $N_{\text{side}} = 1024$ the maximum relative error over every lane falls from 4.2×10^{-2} (plain) to 2.8×10^{-6} (shipped). The identities of the rewrite, a southern-hemisphere reflection onto the same code path, and an exact per-block power-of-two rescale are proved in Lean 4. The kernel is CUDA, called from JAX through the XLA FFI, and is the default latitudinal operator in GMaster, a GPU library with the public API of NaMaster. We benchmark GMaster against widely used state-of-the-art methods and find $20\times$ improvement in end-to-end spin-0 MASTER estimation at $N_{\text{side}} = 1024$, $22\times$ at 2048, and $12\times$ at 8192, relative to NaMaster 3.0 on 96 CPU cores, with bandpowers agreeing to better than 4×10^{-4} at all multipoles on the ACT DR6 scaling board ($N_{\text{side}} = 256\text{--}8192$), at least $17\times$ below the cosmic-variance limit of the bandpowers; the accuracy cost is sub-dominant to survey cosmic variance throughout.

Key words: methods: numerical – cosmology: cosmic microwave background – cosmology: large-scale structure of Universe

1 INTRODUCTION

The MASTER estimator (Hivon et al. 2002) recovers binned angular power spectra from masked spherical maps by analysing the masked fields, forming coupled pseudo-spectra, building a mode-coupling matrix from the mask, and decoupling. Throughout, ℓ and m denote the usual spherical-harmonic multipole and azimuthal order, $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$ are colatitude and azimuth on the sphere, s is the spin of the field, N_{side} is the HEALPix resolution parameter, and $\ell_{\text{max}} = 3N_{\text{side}} - 1$ unless stated otherwise.

NaMaster (Alonso et al. 2019) is the standard CPU implementation of that pipeline: `pymaster` 3.0 exposes `NmtField`, `NmtWorkspace`, and decoupled bandpowers with the normalisations used in current CMB and LSS analyses. Its HEALPix transforms call DUCC0 (Reinecke & Seljebotn 2013), the successor of libsharp, on the HEALPix pixelisation (Górski et al. 2005). The coupling algebra has a structured Wigner- $3j$ recurrence (Kiddier & Gratton 2026). We time NaMaster 3.0 with `ducc0` 0.39.1 on the same host as GMaster.

SHTns (Schaeffer 2013) is a widely used pseudospec-

tral library for on-the-fly associated-Legendre recurrences; release 3.7.5 adds a CUDA implementation (<https://nschaeff.bitbucket.io/shtns/>). Our transform timings use that GPU module (SHT_ALLOW_GPU) from a current source build wired in `benchmarks/scaling/bench_three_way.py`, not the 2013 journal release in isolation. SHTns has no spin-2 transform; spin-2 panels time its spin-1 vector transform as a GPU reference. We do not quote SHTns float32 transform timings: its on-the-fly recurrence is unstable in float32 for the same $1/\sin\phi$ reason as the plain Jacobi march below. GMaster is not benchmarked as a drop-in replacement for arbitrary SHTns grids; the comparison is isolated `alm2map/map2alm` passes at matched ℓ_{max} .

On a GPU the latitudinal stage remains a serial chain in ℓ with a colatitude reduction at every degree—a good match to AVX-512 CPUs, not to SIMT. Precomputed Legendre tables (Price & McEwen 2024) invert the bottleneck when they fit; other GPU spherical-harmonic work (Belkner et al. 2024; Reinecke et al. 2023) is not a drop-in HEALPix MASTER pipeline.

GMaster is a JAX implementation of the NaMaster public API. This Letter isolates one change that made a table-free float32 latitudinal kernel accurate enough to be the default

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at every $N_{\text{side}} \geq 64$: marching the difference pair of the Jacobi row attached to $d_{m,-s}^\ell(\theta)$ rather than the consecutive values. The recurrence, a reflection that sends the southern hemisphere through the same instructions, and the exactness of a per-block power-of-two rescale are proved in Lean 4 (de Moura & Ullrich 2021; The mathlib Community 2020). We report the end-to-end MASTER board and a real-map overlay on ACT DR6. This is not a cosmological analysis: no beam, split cross-spectrum, or covariance is applied to the survey maps.

2 DIFFERENCE-FORM MARCH

Every Wigner function $d_{m,-s}^\ell(\theta)$ is an algebraic multiple of a Jacobi polynomial (Olver et al. 2010, NIST DLMF 14.9.16, 18.3),

$$d_{m,-s}^\ell(\theta) = (-1)^m 2^{\varepsilon_m \text{LG2N}} \times \left(\sin \frac{\theta}{2}\right)^\alpha \left(\cos \frac{\theta}{2}\right)^\beta P_n^{(\alpha,\beta)}(\cos \theta), \quad (1)$$

with $\alpha = m + s$, $\beta = |m - s|$, $n = \ell - \max(m, s)$ and $\text{LG2N} = \log_2 \sqrt{(\ell + m)!(\ell - m)!/((\ell - s)!(\ell + s)!)}$. The family obeys a three-term recurrence in n (Olver et al. 2010, DLMF 18.9.1). Writing $x = \cos \theta$ and v_n for the marched polynomial,

$$v_n = A_n v_{n-1} - B_n v_{n-2}, \quad A_n = c_1 x + c_0, \quad B_n = c_2, \quad (2)$$

where c_1, c_0, c_2 depend only on (ℓ, m) . A float64 march of (2) agrees with a numpy oracle of the same coefficients to 1.5×10^{-15} at $N_{\text{side}} = 1024$. Plain float32 of the same recurrence reaches a maximum relative error of 4.2×10^{-2} over the row (Figure 2, left).

Over a few degrees the coefficients are nearly constant and $c_2 \rightarrow 1$, so the row is locally $v_n \approx A v_{n-1} - v_{n-2}$ with roots $e^{\pm i\phi}$ and $2\sqrt{c_2} \cos \phi = A$. In the oscillatory region $|A| < 2$, and $\phi \rightarrow 0$ at the poles and at turning points $m \approx (\ell + \frac{1}{2}) \sin \theta$. A rounding $\delta \sim 2u|v|$ of $\text{fl}(A v_{k-1} - B v_{k-2})$ is amplified by $1/\sin \phi$ on each mode, giving a relative random-walk floor $\sim \sqrt{n} u / \sin \phi$; on the first HEALPix ring at $N_{\text{side}} = 4096$ this is a few per cent (Figure 2, left). A single-word x is worse: it drives a coherent phase drift $\sim n c_1 x \epsilon / (2 \sin \phi)$.

Let $D_n = v_n - v_{n-1}$. Then for $n \geq 2$

$$D_n = (A_n - 1 - B_n) v_{n-1} + B_n D_{n-1}, \quad v_n = v_{n-1} + D_n. \quad (3)$$

Equation (3) is the central result of this Letter: the latitudinal march is carried in difference form, and every subsequent kernel design choice (two-word C , reflection, rescale) is in service of making that recurrence stable in float32. In terms of ϕ , the difference coefficient is $C = A - 1 - c_2 \approx -\phi^2$, so $D \approx \phi |v|$ exactly where the plain march is ill-conditioned. A rounding of $v \leftarrow \text{fl}(v + D)$ lands on v with D fixed and is not amplified. A rounding of $D \leftarrow \text{fl}(c_2 D + C v)$ lands on D at scale $u\phi|v|$ and, after the $1/(2 \sin \phi)$ modal factor, contributes $O(u|v|)$ per step. Both kinds therefore give a random-walk floor $\sqrt{n} u$ independent of ϕ : $\sqrt{3072} u \approx 3.3 \times 10^{-6}$ at $N_{\text{side}} = 1024$, against the measured 2.8×10^{-6} .

The only coefficient on the amplified path is C , and the argument needs its error to be $O(u|C|)$, not $O(u|A|)$. Writing $C_n = c_1(x - 1) + E_n$ with $E_n = c_1 - 1 - c_2 + c_0$, the kernel carries $x - 1$ rather than x : near a pole $x - 1 \sim -\theta^2/2$, so a single-word x would inject a relative error $u/(\theta^2/2)$ into the leading term of C , and a single-word E would leave an error

$\sim u|c_1|$ that does not shrink with C . Both are systematic and drift the phase. The kernel therefore stores $|x| - 1 = x_h + x_l$, $c_1 = c_{1h} + c_{1l}$ and $E = E_h + E_l$ as float32 pairs, each split from float64 to 2^{-48} (Dekker 1971), and forms

$$C = \text{fma}(c_{1h}, x_h, E_h) + [\text{fma}(c_{1h}, x_l, \text{fma}(c_{1l}, x_h, E_l))]. \quad (4)$$

The first **fma** rounds once, by $u|C|$, and that rounding vanishes with C at the turning point where the amplification peaks; the bracket is $\sim 2^{-24}$ of the leading terms. A single-word C leaves the coherent $u|c_1|$ -class drift, growing like n rather than $\sqrt{n}$. The coefficient c_2 is single-word, because its error enters as $u|D| \sim u\phi|v|$ on D , the unamplified kind.

The southern hemisphere needs no second code path. If $v_n = (c_1 x + c_0) v_{n-1} - c_2 v_{n-2}$ then $w_n := (-1)^n v_n$ satisfies

$$w_n = (c_1(-x) - c_0) w_{n-1} - c_2 w_{n-2}, \quad (5)$$

the same recurrence with $y = |x|$ for x and $-c_0$ for c_0 , whose difference coefficient is $c_1(y - 1) + E_n^-$ with $E_n^- = c_1 - 1 - c_2 - c_0$. Every lane, northern or southern, therefore marches $|x| - 1 \in (-1, 0]$ with the same instructions and one of two table columns $E^\pm$, and the emit restores the sign on odd n . Each lane stores $v \cdot 2^{ex}$ with ex an integer and is rescaled by an exact power of two every eight degrees; the map $(v, D) \mapsto (v + D', D')$ is linear, so the represented value is unchanged, and the per-degree growth is bounded by $1 + |C| + c_2 \leq 2^{15}$, so eight degrees cannot overflow. The emit factorises the Jacobi normalisation $2^{\varepsilon_m \text{LG2N}}$ into a per-lane power of two and a per-degree mantissa proved to lie in $[2^{-45.5}, 2)$, so no exponent lane leaves the kernel. Equations (3) and (5), the seeds, and the rescale are theorems in a Lean 4 development on Mathlib with no **sorry**; the modal error argument is checked by Figure 2 (left).

The kernel is CUDA, one warp per $(m, \text{ring tile})$, exposed as XLA FFI handlers inside **jit**; field and mask share one analysis launch. The default at $L = \ell_{\text{max}} + 1 \geq 192$ ($N_{\text{side}} \geq 64$) is this march; float64 remains available by flag.

3 VALIDATION AND PERFORMANCE

All timings are single-tenant on one NVIDIA RTX PRO 6000 Blackwell (96 GB); NaMaster 3.0 / **ducc0** 0.39.1 on a 96-core Threadripper PRO 9995WX is the CPU reference (Figure 1). End-to-end totals are warmed medians of field construction, coupling matrix, coupled cell and decoupling. Decoupled bandpower agreement $|C_\ell^{\text{GM}}/C_\ell^{\text{NM}} - 1|$ on the ACT DR6 scaling board uses native-resolution maps at each N_{side} (linear bins of 30). After switching mode-coupling Gauss-Legendre weights to **ducc0**, bandpowers agree with NaMaster to better than 4×10^{-4} at all multipoles from $N_{\text{side}} = 256$ to 8192, at least $17\times$ below cosmic variance; at $N_{\text{side}} = 8192$ TT maxima fall from 7.1×10^{-3} to 3.8×10^{-5} and EE from 7.3×10^{-3} to 2.8×10^{-4} (Figure 2). On the ACT coadd at $N_{\text{side}} = 4096$ (Figure 3), decoupled TT has rms($C_\ell^{\text{GM}}/C_\ell^{\text{NM}} - 1$) = 1.41×10^{-5} in 7.63s versus 70.4s on 192 cores.

4 CONCLUSIONS

A three-term Jacobi recurrence in float32 is limited by $1/\sin \phi$ amplification, not by the width of its words. Marching $(v, v - v_{\text{prev}})$ moves the injection of every rounding onto a difference

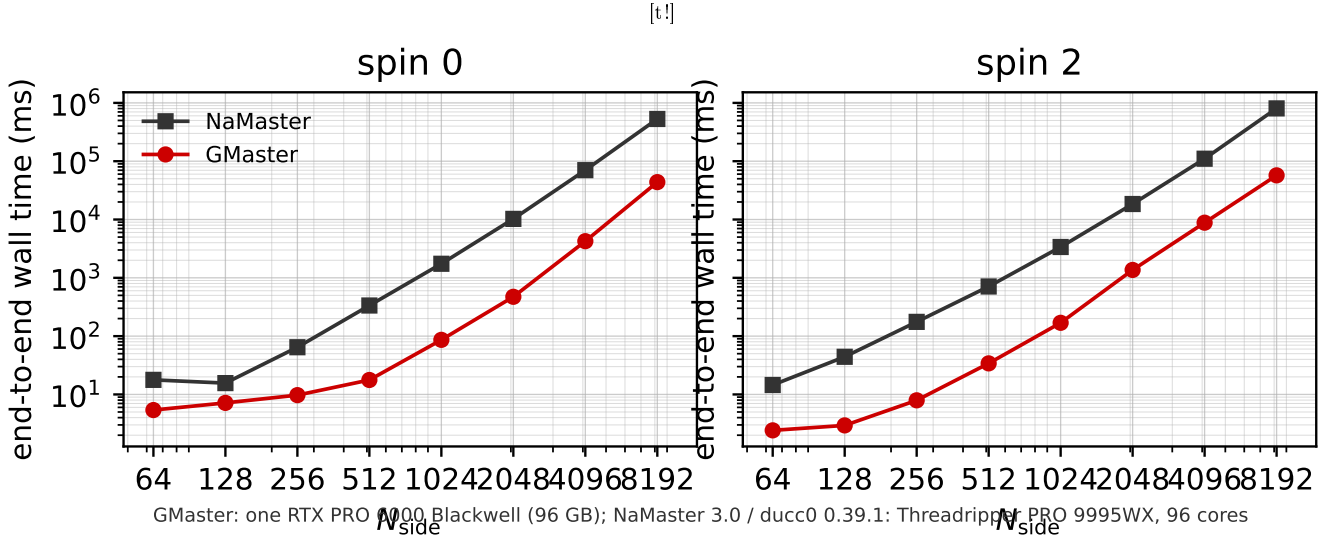


Figure 1. End-to-end MASTER wall time on the host named in the figure footer. Grey: NaMaster on 96 cores; red: GMaster on one RTX PRO 6000 Blackwell. Speed-ups are 12–22× for spin 0 and 14–22× for spin 2 over $N_{\text{side}} = 1024$ –8192.

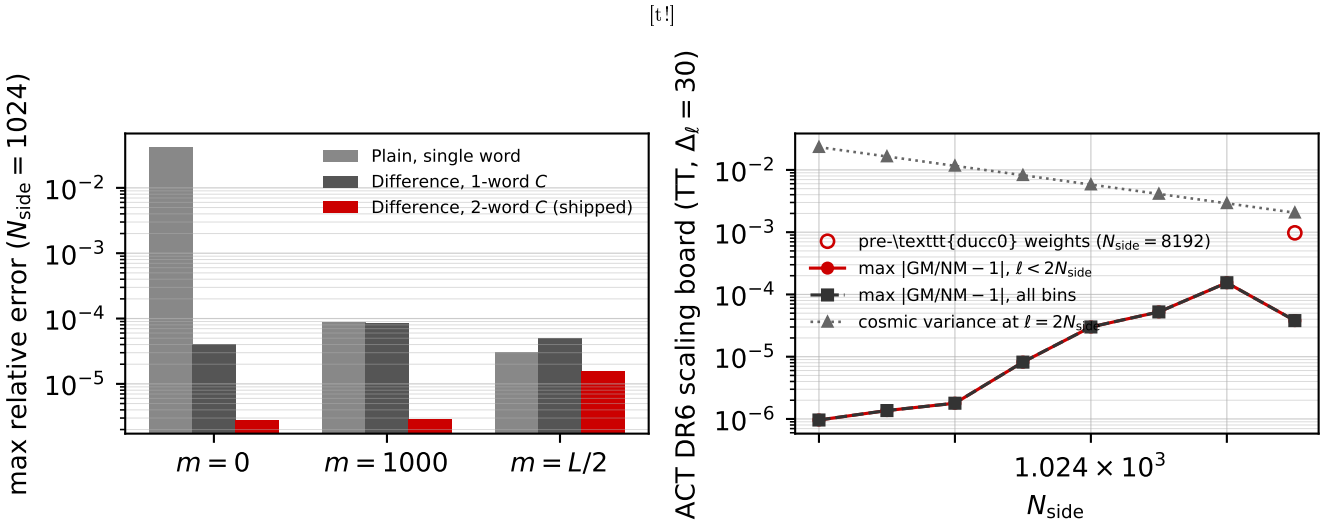


Figure 2. Left: maximum relative error of a float32 march against float64, over every lane at $N_{\text{side}} = 1024$ spin 0 ($m = 0, 1000, L/2$): plain recurrence (grey) versus difference form with two-word C (red, shipped). Right: maximum $|C_{\ell}^{\text{GM}}/C_{\ell}^{\text{NM}} - 1|$ on decoupled TT bandpowers from the ACT DR6 end-to-end scaling board ($f_{\text{sky}} = 0.481$, $\Delta_{\ell} = 30$); solid red: bins with $\ell < 2N_{\text{side}}$, dashed grey: all bins; dotted grey: cosmic variance at $\ell = 2N_{\text{side}}$. The hollow marker at $N_{\text{side}} = 8192$ is the pre-`ducc0`-weight residual; filled markers use post-switch bandpowers (3.8×10^{-5} at 8192).

that already vanishes at the turning points, and a two-word C removes the residual coherent drift. Carried as a CUDA kernel behind NaMaster's API, that rewrite is the default latitudinal operator in GMaster and is 10–22× NaMaster at $N_{\text{side}} = 128$ –8192 on the board of Figure 1. The same pipeline on an ACT DR6 coadd at $N_{\text{side}} = 4096$ agrees at 10^{-5} in under nine seconds.

No other GPU spherical-harmonic implementation was timed on the full MASTER board. The shipped default trades NaMaster's float64 transform contract for decoupled bandpowers that match NaMaster to better than 4×10^{-4} at all

multipoles on the ACT scaling board ($N_{\text{side}} = 256$ –8192), at least 17× below cosmic variance; the exact float64 route is one flag. The overlay is a methods comparison, not a science product.

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NaMaster, DUCC/libsharp, s2fft, threej_cosmo and SHTns are the reference stack against which the numbers were measured. The Lean 4 development uses Mathlib. ACT DR6

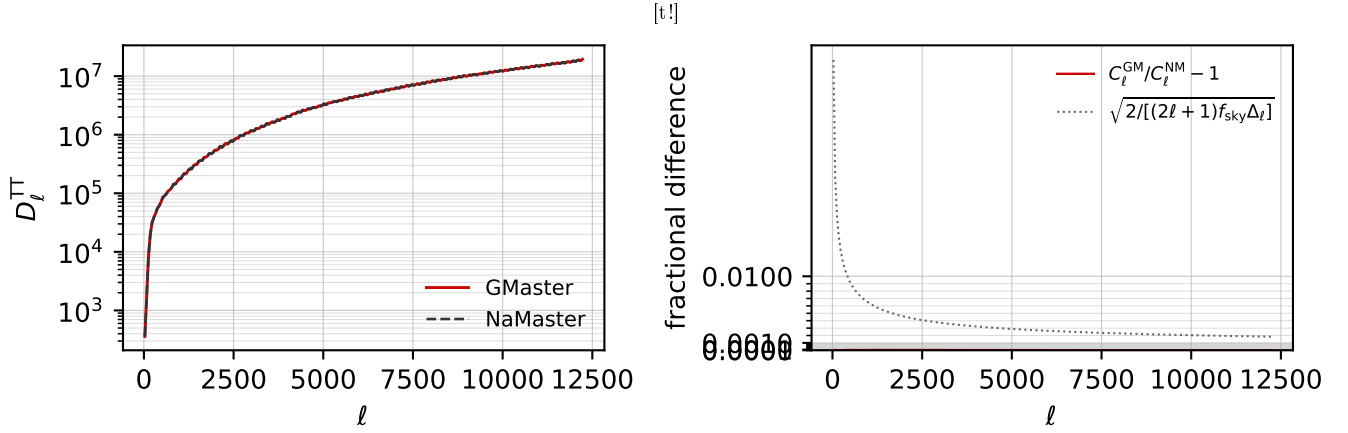


Figure 3. ACT DR6 night PA4 f_{150} source-free coadd (Naess et al. 2025) at $N_{\text{side}} = 4096$ ($f_{\text{sky}} = 0.481$). Left: decoupled TT D_ℓ after the full MASTER chain (NmtField with $n_{\text{iter}} = 3$ mask refinement, coupling matrix, coupled cell, decoupling; linear bins of width 50); GMaster (red) and NaMaster (grey) share inputs. GMaster executes the latitudinal march of Section 2 inside every spin-0 analysis/synthesis that builds field and mask alms. Right: fractional bandpower difference $C_\ell^{\text{GM}}/C_\ell^{\text{NM}} - 1$ with the cosmic-variance envelope $\sqrt{2/[(2\ell + 1)f_{\text{sky}}\Delta\ell]}$ (dotted). Methods overlay, not an ACT science product.

maps are those of Naess et al. (2025). Numerical claims were measured on the hardware of Section 3; parts of the library and formal proof were developed with coding-agent assistance under author supervision.

DATA AVAILABILITY

GMaster, the overlay scripts, the committed bandpowers of the ACT overlay and a notebook that rebuilds Figure 3 and the timing figure from them are at <https://github.com/licongxu/GMaster>. Figure 3 uses the public ACT DR6 maps (Naess et al. 2025).

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